SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1701**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

*I _**Shaik Thouhid**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
thouhid

Annexure A
Dry Run Evaluation

1. Question 1: Breadth-First Search (BFS)

Graph (tree): A is the root with children B and C; B has children D and E; C has children F and G. Children are visited left to right.

Method: BFS uses a FIFO queue. Remove the front node, mark it visited, and add its children to the back of the queue. The queue column shows the contents after the node is visited and expanded.

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	(empty)	G

Answers

1. BFS traversal order: **A → B → C → D → E → F → G**.
2. Queue contents after each iteration: [B, C], [C, D, E], [D, E, F, G], [E, F, G], [F, G], [G], [ ] (initially the queue held only [A]).

2. Depth-First Search (DFS)

Same graph as above. DFS uses a LIFO stack. Pop the top node, mark it visited, and push its children so that the left child is on top (the right child is pushed first). The top of the stack is written on the left.

Iteration	Stack (top on left)	Visited Node
1	B, C	A
2	D, E, C	B
3	E, C	D
4	C	E
5	F, G	C
6	G	F

Iteration	Stack (top on left)	Visited Node
7	(empty)	G

Answer

1. DFS traversal order: **A → B → D → E → C → F → G**.

DFS goes as deep as possible along one branch (A, B, D) before backtracking to the next unexplored branch (E, then C and its subtree), whereas BFS finishes each level first.

3. BFS on the 8-Puzzle

Initial state and goal state (0 denotes the blank tile):

Initial State	Goal State
1 3 6 5 0 2 4 7 8	1 2 3 4 5 6 7 8 0

Assumptions: the blank is moved in the order Up, Down, Left, Right; a state already generated is not generated again; the goal test is applied when a state is removed from the queue. States are labelled S0, S1, S2, ... in the order they are generated (S0 is the initial state).

BFS trace (first five iterations)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	**S0** 1 3 6 5 0 2 4 7 8	S0	S1, S2, S3, S4
2	**S1** 1 0 6 5 3 2 4 7 8	S1, S2, S3, S4	S2, S3, S4, S5, S6
3	**S2** 1 3 6 5 7 2 4 0 8	S2, S3, S4, S5, S6	S3, S4, S5, S6, S7, S8
4	**S3** 1 3 6 0 5 2 4 7 8	S3, S4, S5, S6, S7, S8	S4, S5, S6, S7, S8, S9, S10
5	**S4** 1 3 6 5 2 0 4 7 8	S4, S5, S6, S7, S8, S9, S10	S5, S6, S7, S8, S9, S10, S11, S12

States generated in these iterations (S0 expands to S1 to S4; S1 to S5, S6; S2 to S7, S8; S3 to S9, S10; S4 to S11, S12):

S1 (S0, Up)	**S2 (S0, Down)**	**S3 (S0, Left)**	**S4 (S0, Right)**
1 0 6 5 3 2 4 7 8	1 3 6 5 7 2 4 0 8	1 3 6 0 5 2 4 7 8	1 3 6 5 2 0 4 7 8
S5 (S1, Left)	**S6 (S1, Right)**	**S7 (S2, Left)**	**S8 (S2, Right)**
0 1 6 5 3 2 4 7 8	1 6 0 5 3 2 4 7 8	1 3 6 5 7 2 0 4 8	1 3 6 5 7 2 4 8 0
S9 (S3, Up)	**S10 (S3, Down)**	**S11 (S4, Up)**	**S12 (S4, Down)**
0 3 6 1 5 2 4 7 8	1 3 6 4 5 2 0 7 8	1 3 0 5 2 6 4 7 8	1 3 6 5 2 8 4 7 0

The queue grows by one state per iteration at the start, and BFS continues in the same way, level by level (depth 0, then 1, then 2, and so on), until the goal state is removed from the queue. The goal lies at depth 8.

Answers

1. Which node is expanded in each iteration? Nodes are expanded in the order they entered the queue (FIFO): S0, S1, S2, S3, S4 in iterations 1 to 5, then S5, S6, S7, ... and so on. All depth-1 states are expanded before any depth-2 state, and so on, until the goal is dequeued at iteration 311.
2. How many states are generated in total? **500 states** (including the initial state), under the assumptions above. Of these, 310 states were expanded and the goal state was the 311th state dequeued. The exact total depends on the move order and on whether the goal test is made at generation or at expansion.
3. Complete solution path (8 moves of the blank tile: Right, Up, Left, Down, Left, Down, Right, Right):

Step 0: Start	**Step 1: Right**	**Step 2: Up**
1 3 6 5 0 2 4 7 8	1 3 6 5 2 0 4 7 8	1 3 0 5 2 6 4 7 8
Step 3: Left	**Step 4: Down**	**Step 5: Left**
1 0 3 5 2 6 4 7 8	1 2 3 5 0 6 4 7 8	1 2 3 0 5 6 4 7 8

Step 6: Down	Step 7: Right	Step 8: Right (Goal)
1 2 3 4 5 6 0 7 8	1 2 3 4 5 6 7 0 8	1 2 3 4 5 6 7 8 0

(The move shown is the direction in which the blank tile moves.)

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle? In the 8-puzzle every move has the same cost (one step). BFS expands states strictly in order of increasing depth: every state at depth d is expanded before any state at depth d + 1. Hence the first time the goal is generated or removed from the queue, it has been reached by a path with the smallest possible number of moves, because no shallower path to the goal exists (otherwise the goal would already have been found at an earlier level). Since the number of moves from any state is finite (at most four), BFS is also complete, so it always finds a solution when one exists. The price is memory: it must store every state of the current frontier.